Sankalp Arun Karkera

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Professional Summary

Results-driven Data Analyst with 7+ years of experience architecting and delivering enterprise-scale data solutions for Fortune 500 organizations. Proven ability to analyze petabyte-scale datasets and transforming complex analytical insights into actionable business strategies that drive measurable revenue growth and operational efficiency. Expert in building end-to-end cloud-based data pipelines, implementing advanced analytics frameworks, and establishing robust data governance standards across cross-functional teams.

Work Experience

Data Analyst — Marriott International (TCS) 2019 – Present

MARSHA Revenue Intelligence Platform (Marriott Automated Revenue Strategy & Hospitality Analytics)

- Built revenue analytics workflows processing 50M+ booking records from Snowflake data warehouse to predict optimal room rates based on 200+ features including seasonality, events, weather, and competitor pricing.
- Conducted in-depth exploratory data analysis using Python to uncover trends in guest behavior, booking patterns, and seasonal occupancy across 8000+ properties, identifying key demand drivers that improved forecasting accuracy by 15%.
- Transformed raw data into insightful reports using Python and SQL, enabling the business to track key KPIs like RevPAR, ADR, and guest satisfaction scores through automated daily reporting pipelines integrated with Snowflake and SQL Server.
- Created materialized views in Snowflake to pre-aggregate revenue performance metrics, reducing dashboard load times by 40% and enabling real-time pricing decisions across regional property portfolios.
- Developed interactive dashboards in Tableau connecting directly to Snowflake, featuring drill-down capabilities from portfolio level to individual property ADR performance, with automated alerts triggering when pricing deviates beyond defined thresholds.

Bonvoy CDP Enhancement (Customer Data Platform Analytics)

- Engineered comprehensive customer data model in Snowflake using complex SQL queries to calculate RFM scores, lifetime value, and behavioral segmentation across 180M+ Bonvoy member records.
- Categorized guest profiles and loyalty membership tiers, enabling faster lookups to enrich data for personalized marketing insights and reducing query response times by 60%.
- Collaborated with engineering and product teams to validate and clean data before analysis, using SQL for data wrangling and Python for rule-based data quality checks, ensuring 99.5% data accuracy across customer touchpoints.
- Built interactive Power BI dashboards for customer acquisition cost, churn prediction probabilities, and segment migration analysis, enabling marketing teams to create targeted campaigns worth $12M in incremental revenue.

MMP Attribution Engine (Marriott Marketing Platform Analytics)

- Implemented multi-touch attribution models using Python and SQL to analyze customer journey data from Adobe Analytics, and internal booking systems, processing 10M+ monthly digital touchpoints to optimize marketing spend allocation.
- Analyzed campaign performance data to identify trends, inefficiencies, and growth opportunities across marketing channels, resulting in optimized targeting strategies that improved ROI by 35% across digital advertising spend.
- Created and maintained Python-based reporting pipelines integrated with Snowflake and SQL Server, ensuring consistent and accurate data feeds for dashboards tracking campaign performance across 50+ marketing channels.
- Translated campaign insights into actionable recommendations to refine targeting and budget allocation, leading to more effective marketing spend distribution across digital channels and $8M in cost savings annually.
- Created pre-aggregated campaign performance metrics in Snowflake to parse, extract, and transform nested data from advertising platforms, reducing dashboard refresh times from 45 minutes to 3 minutes.

MICROS Business Intelligence Extension (Market Intelligence & Competitive Reporting Optimization System)

- Conducted in-depth data analysis using Python to uncover competitive positioning trends, market share shifts, and pricing elasticity patterns across key metropolitan markets.

- Developed sophisticated SQL queries using analytical functions and pivot operations to calculate market share, rate positioning, and demand transfer analysis, creating stored procedures for automated monthly competitive intelligence reporting.
- Transformed raw competitive data into insightful reports using Python and SQL, enabling revenue management teams to track market penetration indices and competitive positioning across 150+ markets.
- Collaborated with engineering and product teams to validate and clean operational data before analysis, implementing automated data quality checks using Python that identified and resolved 95% of data inconsistencies before reporting.
- Constructed Tableau dashboards with advanced calculations including market penetration indices, competitive positioning heat maps, and demand elasticity visualizations, featuring parameter controls for dynamic market selection and time period analysis.

MGS Analytics Platform (Marriott Guest Services Sentiment Intelligence)

- Analyzed 2M+ guest reviews & feedbacks across multiple platforms, implementing sentiment scoring, topic modeling & trend analysis.
- Transformed unstructured feedback data into actionable insights using Python and SQL, creating automated reporting systems that track guest satisfaction KPIs and operational impact metrics in real-time.
- Created Monthly deck dashboards in Power BI and ServiceNow to provide visual representations of sentiment analysis, service recovery metrics, and operational correlation data, improving guest experience management across 500+ properties.
- Presented analytical findings to operations and guest services leadership, translating sentiment analysis results into operational improvements that reduced negative review response time by 75% and improved overall satisfaction scores by 22%.

Software Engineer — Infogain Corporation 2016 – 17

Enterprise Keyword Engine Integration

- Conducted extensive database tuning and optimization initiatives including index restructuring, query plan analysis, and stored procedure optimization, resulting in a 35% improvement in overall database performance across production environments.
- Managed complete application development lifecycle from requirements gathering through production deployment, creating functional specifications, technical documentation, and coordinating with cross-functional teams for seamless integration with existing enterprise systems.
- Developed automated monitoring and alerting systems using SQL Server Agent jobs and custom scripts to track data pipeline performance, proactively identifying bottlenecks and ensuring 24/7 system availability for critical business operations.
- Optimized database queries through advanced techniques including query rewriting, partition pruning, and materialized view implementation, achieving a 40% decrease in query response time for high-frequency reporting queries used by executive dashboards.

Education

Master of Science in ***Information Systems*** **—** New Jersey Institute of Technology **2019**

Bachelor of Engineering in ***Computer Science*** **—** University of Mumbai **2015**

Projects

General Purpose AI Agent - GAIA Benchmark Evaluation

- Created custom general-purpose AI agent capable of handling complex, multi-step reasoning tasks and evaluated its performance using a subset of the GAIA (General AI Assistant) benchmark dataset to measure real-world problem-solving capabilities.
- Built multi-modal agent system using Hugging Face Agents framework with tool integration for web search, code execution, mathematical computation, and file processing, designed to tackle GAIA's diverse question types requiring advanced reasoning and external knowledge access.
- Evaluated agent performance across 200+ GAIA benchmark questions spanning multiple difficulty levels and categories (factual, computational, reasoning, multimodal), implementing automated scoring system to measure accuracy, reasoning quality, and task completion rates, there by achieving 73% accuracy on GAIA Level 1 questions, 58% on Level 2, and 45 % on Level 3, with detailed analysis identifying strengths in factual retrieval and computational tasks.

Skills

Languages & Databases: Python, SQL, PostgreSQL, MS SQL Server, Netezza, IBM DB2, Oracle

Cloud Services & Gen AI: AWS, S3, EC2, AWS Certified, Agent AI Dev, Smolagents, LangChain, LangGraph

Big Data & Modelling Tools: Tableau, Power BI, Snowflake, Airflow

Environment & frameworks: Jenkins, UNIX, Putty, ServiceNow